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Electromagnetic Energy Exchange in Vacuum

charge conservation:

$$\frac{\partial}{\partial t} \varrho + \nabla \cdot j = 0 \quad (1)$$

$\varrho = \sigma \delta = \varepsilon_0 \nabla \cdot \mathcal{E} =$ charge density in interior

$\sigma =$ charge density at border (skin)

$\delta =$ Dirac delta-function

$j = \sigma (\mathcal{E} + v \times \mathcal{B}) = \nabla \times \mathcal{H} - \frac{\partial}{\partial t} \mathcal{D} = \frac{1}{\mu_0} \nabla \times \mathcal{B} - \varepsilon_0 \frac{\partial}{\partial t} \mathcal{E} \hat{=}$ electric current density

$j \cdot \mathcal{E} \hat{=}$ energy gain by fields $\mathcal{E}$ and $\mathcal{B}$ through current density j

$\mathcal{S} = \mathcal{E} \times \mathcal{H} \hat{=}$ energy transport density (Intensity = $|\overline{\mathcal{S}}|$)

Energy conservation for (large extended) conductive medium:

$$\int \left(\frac{\partial}{\partial t} (w_e + w_m) + \nabla \cdot \mathcal{S} + j \cdot \mathcal{E} \right) dr = \frac{d}{dt} W_e + \frac{d}{dt} W_m + D_S + \int (j \cdot \mathcal{E}) dr \stackrel{\text{low frequency}}{=} \frac{1}{2} \frac{d}{dt} (LI^2) + \frac{1}{2} \frac{d}{dt} \left(\frac{Q^2}{C} \right) \quad (2)$$

Energy Balance

Electro-Magnetic Energy:

$$E = E_{kin} + E_{pot} = \sqrt{(pc)^2 + (mc^2)^2} \stackrel{\text{vacuum}}{=} pc = \hbar\omega = \text{chargeenergy} + \text{fieldenergy} + \text{transportenergy} \quad (3)$$

Charge Energy:

$$\underbrace{j \cdot \mathcal{E}}_{\text{charge energy}} = - \left(\underbrace{\frac{\partial}{\partial t} \sum w_i}_{\text{field energy}} + \underbrace{\nabla \cdot \mathcal{S}}_{\text{transfer energy}} \right) = \mathcal{E} \cdot \nabla \times \mathcal{A} - \mathcal{E} \cdot \frac{\partial}{\partial t} \mathcal{D} \quad (4)$$

Field Energy:

$$\frac{\partial}{\partial t} w = \frac{\partial}{\partial t} \sum w_i = \frac{\partial}{\partial t} (w_e + w_m) = \frac{\partial}{\partial t} \left(\frac{\varepsilon_0}{2} \mathcal{E}^2 + \frac{1}{2\mu_0} \mathcal{B}^2 \right) \stackrel{\text{vacuum}}{=} \frac{\partial}{\partial t} \left(\frac{1}{2} \frac{1}{\mu_0} (\mathcal{A} \cdot \nabla \times \mathcal{B} - \mathcal{B}^2) \right) \quad (5)$$

Transfer Energy:

$$- \dot{w} \hat{=} \nabla \cdot \mathcal{S} \stackrel{\text{matter}}{=} \nabla \cdot (\mathcal{E} \times \mathcal{H}) \stackrel{\text{vacuum}}{=} \nabla \cdot (\mathcal{E} \times \mathcal{B}) \stackrel{\text{alternativ in vacuum}}{=} \nabla \cdot \left(\frac{1}{2} \frac{1}{\mu_0} \frac{\partial}{\partial t} (\mathcal{A} \times \mathcal{B}) \right) = \dots \quad (6)$$

Local Energy Conservation:

$$\text{Force } F = q(\mathcal{E} + v \times \mathcal{B})$$

$$\text{Work } W = \Delta E = Fr = pv = \text{const.}$$

$$\text{Power } \frac{d}{dt} W = 0$$

$$\overbrace{\underbrace{j \cdot \mathcal{E}}_{\text{charge energy}} + \underbrace{\frac{\partial}{\partial t} \sum w_i}_{\text{field energy}} + \underbrace{\nabla \cdot \mathcal{S}}_{\text{transfer energy}}}_{= \text{const.}} \stackrel{\text{alternatives}}{=} j \cdot \mathcal{E} + \frac{\partial}{\partial t} \left(\frac{1}{2} \varepsilon_0 \mathcal{E}^2 + \frac{1}{2} \frac{1}{\mu_0} \mathcal{A} \cdot \nabla \times \mathcal{B} \right) + \frac{1}{\mu_0} \nabla \cdot \left(\mathcal{E} \times \mathcal{B} + \frac{1}{2} \frac{\partial}{\partial t} \right) \quad (7)$$

conditions for modeling alternatives (Eigenvalue solution):

- $w = \sum w_i : \frac{\partial}{\partial t} w = -\nabla \cdot \mathcal{S}_w$
- $w = w(\mathcal{E}, \mathcal{B}) : \mathcal{E} \lambda^2 = \varrho \lambda$
- $w = w(\mathcal{E}, \mathcal{B}) : \mathcal{B} \lambda^2 = j \lambda$
- $\mathcal{S}_w = \mathcal{S}_w(\mathcal{E}, \mathcal{B}) : \mathcal{E} \lambda^2 = \varrho \lambda$
- $\mathcal{S}_w = \mathcal{S}_w(\mathcal{E}, \mathcal{B}) : \mathcal{B} \lambda^2 = j \lambda$

Energy Density Flux: Poynting-Vector

$$\mathcal{S} \stackrel{vacuum}{=} \mathcal{E} \times \mathcal{B} \stackrel{matter}{=} \mathcal{E} \times \mathcal{H}$$

Far Field: Power = $\frac{\Delta E}{t} \hat{=} \frac{d}{dt} E = 0 \Rightarrow \text{Energy} = \text{const.} \Rightarrow \text{Energy Conservation}$

$$\frac{d}{dt} E \hat{=} \frac{d}{dt} \left(\underbrace{\int_{\mathbb{R}^3} j \cdot \mathcal{E}}_{\text{charge energy}} + \underbrace{\int_{\mathbb{R}^3} \frac{\partial}{\partial t} \sum w_i}_{\text{field energy}} \right) = 0$$

= const.

Near Field (local): Power = Energytransport

$$\frac{d}{dt} E \hat{=} \frac{d}{dt} \left(\underbrace{\int_{\mathbb{R}^3} j \cdot \mathcal{E}}_{\text{charge energy}} + \underbrace{\int_{\mathbb{R}^3} \frac{\partial}{\partial t} \sum w_i}_{\text{field energy}} \right) \stackrel{local}{=} \underbrace{- \int_{\mathbb{R}^2} \mathcal{S} \cdot \mathbf{n} \, df}_{\text{transported energy density}} \stackrel{\text{"skin"}}{=}$$

Local Force Balance

Electro-Magnetic Force:

$$\dot{p} = F = q(\mathcal{E} + v \times \mathcal{B}) = \text{chargeforce} + \text{fieldforce} + \text{stresstension} \quad (8)$$

Charge Force:

$$\dots = \rho \mathcal{E} + j \times \mathcal{B} \quad (9)$$

Field Force:

$$\dots = \frac{\partial}{\partial t} \frac{1}{c^2} \mathcal{S} \quad (10)$$

Tension Force:

$$\nabla \cdot \mathcal{T} = \varepsilon_0 (\mathcal{E} \times \nabla \times \mathcal{E} - \mathcal{E} \nabla \cdot \mathcal{E}) + \frac{1}{\mu_0} (\mathcal{B} \nabla \times \mathcal{B} - \mathcal{B} \nabla \cdot \mathcal{B}) \quad (11)$$

Local Momentum Conservation:

$$\text{Force } F = q(\mathcal{E} + v \times \mathcal{B}) = \frac{d}{dt} p = 0$$

$$\text{Momentum } p = \int F = \frac{E}{c} = \text{const.}$$

$$\overbrace{\underbrace{\rho \mathcal{E} + j \times \mathcal{B}}_{\text{charge force}} + \underbrace{\frac{\partial}{\partial t} \frac{1}{c^2} \mathcal{S}}_{\text{field force}} + \underbrace{\nabla \cdot \mathcal{T}}_{\text{tension force}}}^{\dots} = 0 \quad (12)$$

Tension Force: Stress-Tensor

Stress Tensor quantifies tension $\hat{=}$ Forces = Pressure - Traction $\hat{=}$ “torsion tension” at angular momentum

$$\mathcal{T} \hat{=} \mathcal{T}_{ik} = \delta_{ik} \left(\frac{1}{2} \varepsilon_0 \mathcal{E}^2 + \frac{1}{2} \frac{1}{\mu_0} \mathcal{B}^2 \right) - \varepsilon_0 \mathcal{E}_i \mathcal{E}_k - \frac{1}{\mu_0} \mathcal{B}_i \mathcal{B}_k \quad (13)$$

$\mathcal{E}$ and $\mathcal{B}$ fields of spacial confined charge and current density distribution, far field, static:
 $\mathcal{T} \sim \frac{1}{r^4} \Leftrightarrow \int_{\mathbb{R}^3} \mathcal{T} d^3r = \int_{\infty} \mathcal{T} df = 0$

Far Field: Force = 0 $\Leftrightarrow$ Momentum = const. $\Rightarrow$ Momentum Conservation

$$F = \dot{p} \hat{=} \frac{d}{dt} \underbrace{\left[\underbrace{\int_{local} (\varrho \mathcal{E} + j \times \mathcal{B})}_{\text{charge momentum}} + \underbrace{\frac{1}{c^2} \mathcal{S}}_{\text{field momentum}} \right]}_{=p=const.} = 0$$

Near Field (local): Force = Tension

$$F = \dot{p} \hat{=} \frac{d}{dt} \underbrace{\left[\underbrace{\int_{local} (\varrho \mathcal{E} + j \times \mathcal{B})}_{\text{charge momentum}} + \underbrace{\frac{1}{c^2} \mathcal{S}}_{\text{field momentum}} \right]}_{=p \neq const., \exists -\mathcal{T} \text{ local tension}} \stackrel{local}{=} - \underbrace{\int_{\mathbb{R}^2} \mathcal{T} \cdot \mathbf{n} df}_{\text{stress tension appears "skin"}}$$

Momentum Balance

Momentum Conservation and Local Tension: linear momentum conservation if $p = const.$, otherwise “tension” occurs:

$$p = \int F = \int \frac{d}{dt} p = \sqrt{\frac{1}{c^2} (E^2 - (mc)^2)} \stackrel{\text{vacuum}}{=} \frac{E}{c} = \frac{2\pi}{c^2} h\nu$$

$p = const.$, linear, vacuum, Far Field:

$$F = \dot{p} \hat{=} \int_{\mathbb{R}^3} \frac{d}{dt} \left[\underbrace{(\varrho \mathcal{E} + j \times \mathcal{B})}_{\text{charge force density}} + \underbrace{\frac{\partial}{\partial t} \frac{1}{c^2} \mathcal{S}}_{\text{field force density}} \right] = \frac{d}{dt} \left[\underbrace{\int_{\mathbb{R}^3} (\varrho \mathcal{E} + j \times \mathcal{B})}_{\text{charge momentum}} + \underbrace{\frac{1}{c^2} \mathcal{S}}_{\text{field momentum}} \right] = 0$$

= p momentum conserved if p = const. (14)

$p \neq \text{const.}$, local $\in \mathbb{R}^2 \ll \mathbb{R}^3 = \infty$, linear, vacuum, Near Field $\Leftrightarrow \exists -\mathcal{T}$ as tension compensation flux:

$$F = \dot{p} \hat{=} \int_{\text{local}} \frac{d}{dt} \left[\underbrace{(\varrho \mathcal{E} + j \times \mathcal{B})}_{\text{charge force density}} + \underbrace{\frac{\partial}{\partial t} \frac{1}{c^2} \mathcal{S}}_{\text{field force density}} \right] = \frac{d}{dt} \left[\underbrace{\int_{\text{local}} (\varrho \mathcal{E} + j \times \mathcal{B})}_{\text{charge momentum}} + \underbrace{\frac{1}{c^2} \mathcal{S}}_{\text{field momentum}} \right] \stackrel{\text{local}}{=} - \int_{\mathbb{R}^2} \mathcal{T} \cdot \mathbf{n} \underbrace{d}_{\text{"skin"}}$$

= $p \neq \text{const.}$, $\exists -\mathcal{T}$ local tension (15)

Momentum $p = \int_{\mathbb{R}^3} (\varrho \mathcal{E} + j \times \mathcal{B}) + \frac{1}{c^2} \mathcal{S}$

photon Spin $= \int_{\mathbb{R}^3} \frac{1}{c^2} \mathcal{S}$

angular momentum (photon Spin):

$$D = \dot{L} \hat{=} \dots \quad (16)$$

Radiation Pressure

average energy density $\hat{=}$ radiation pressure, if stationary (very slow flow motion $v \ll c$) :

$$\overline{w} \hat{=} -\nabla \mathcal{T} \cdot \mathbf{n} \hat{=} -\frac{1}{c} |\overline{\mathcal{E} \times \mathcal{H}}| = -\frac{1}{c} |\overline{\mathcal{S}}| = \overline{(\varrho \mathcal{E} + j \times \mathcal{B}) + \frac{\partial}{\partial t} \frac{1}{c^2} \mathcal{S}} \stackrel{\text{stationary}}{=} \overline{\varrho \mathcal{E} + j \times \mathcal{B}} \quad (17)$$

Abbreviations:

w_i = energy densities (electric w_e , magnetic w_m , gravitational, etc. ...), $i \in \mathbb{N}$

$\mathcal{S}_{w_i}$ = Poynting-Vectors $\hat{=}$ energy transport densities (transfers) by fields (electric, magnetic, gravitational, etc. ...) to corresponding energy densities w_i

$j \cdot \mathcal{E}$ = energy gain provided by fields $\mathcal{E}$ and $\mathcal{B}$ through current density j

$\mathcal{T}$ = Stress Tensor, (tension, momentum flow density)

$\mathbf{n}$ = surface normal vector (orthogonal on skin)

$\Delta \cdot := \text{div grad} = \text{Laplacian}$

J = “ignition of inhomogeneity, cause”

j = Current density

ϱ = Charge density

V = Potential difference

I = Current

R = Resistance

L = Inductance

C = Capacitance

Q = charge (ion)

$\Psi = LI$ = magn. Flux

$\mathcal{S} = \mathcal{E} \times \mathcal{H}$ = Poynting-Vector, radiation intensity

$D_{\mathcal{S}} = \int (\cdot \mathcal{S}) dr = \int \text{div}(\mathcal{E} \times \mathcal{H}) dr$ = dissipation energy lost from radiation ≈ 0 for low frequencies ω

$d = \sqrt{\frac{1}{\mu\sigma\omega}} = \text{skin depth (surface) of conductor}$

$r = \frac{d}{\sqrt{2}i}\rho = \frac{d}{1+i}\rho$ = radius (extension) of conductor

$\rho = kr$ = “radius of curvature k ”

μ = permeability of conductor

σ = conductivity of conductor (dielectricum)